PROJECT TITLE – File Integrity Monitoring System Using Cryptographic Hashing.

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Synopsis Report Preparation

INTRODUCTION

- In the digital era, maintaining file security and data integrity is very important. Files can be modified due to unauthorized access, malware, or human error.
- A File Integrity Monitoring System (FIMS) helps detect such changes by continuously monitoring files. It uses cryptographic hashing techniques like SHA-256 to generate a unique hash value for each file.
- If any change occurs in the file, its hash value also changes, allowing the system to quickly identify tampering and alert the user.
- This project focuses on building a simple and efficient system to ensure file integrity and improve data security.

PROBLEM STATEMENTS:

The **File Integrity Monitoring System using Cryptographic Hashing** is designed to ensure

the security and integrity of files by detecting any unauthorized changes. The system focuses on monitoring selected files or directories and verifying their integrity using hashing techniques.

It specifies :

- Monitoring important files and folders for any changes such as modification, deletion, or addition.
- Generating **cryptographic hash values** (e.g., SHA-256) for files and storing them securely.
- Comparing current hash values with previously stored values to detect tampering.
- Providing alerts or notifications when any unauthorized change is detected.

- Maintaining logs of file activities for auditing purposes.

Literature Review:

- Existing research focuses on file security using cryptographic hashing (MD5, SHA-256).
- File Integrity Monitoring (FIM) tools like Tripwire detect unauthorized changes.
- Hash values help identify even minor file modifications.

Logging and auditing are used to track file activities and security breaches.

Research Gap:

- Need for a simple, lightweight, and user-friendly system.
- Efficient logging with low resource usage.

Objectives of the system:

- To enhance data security by detecting unauthorized modifications
- To ensure data integrity using reliable hashing algorithms.
- To provide a simple and efficient monitoring solution for users.

Target users / system limits:

- Individual users, system administrators, and small organizations.
- Limited to monitoring files on a local system (or defined network scope if extended).

Methodology :

- **Data Collection:** Select important files/folders to monitor.

- **Tools & Technologies:** Python/Java, hashing algorithms (SHA-256), file handling.
- **Algorithm Used:** Generate hash → store baseline → compare with current hash.
- **Process:**

  Scan files

  Detect changes

  Generate alerts & maintain log

Testing & Validation: Test by modifying files and verify detection accuracy and log record.

Technologies / tools used :

- Programming language (e.g., Python / Java)
- Cryptographic hashing algorithms (e.g., SHA-256, MD5)
- File handling and logging mechanisms

Constraints and limitations:

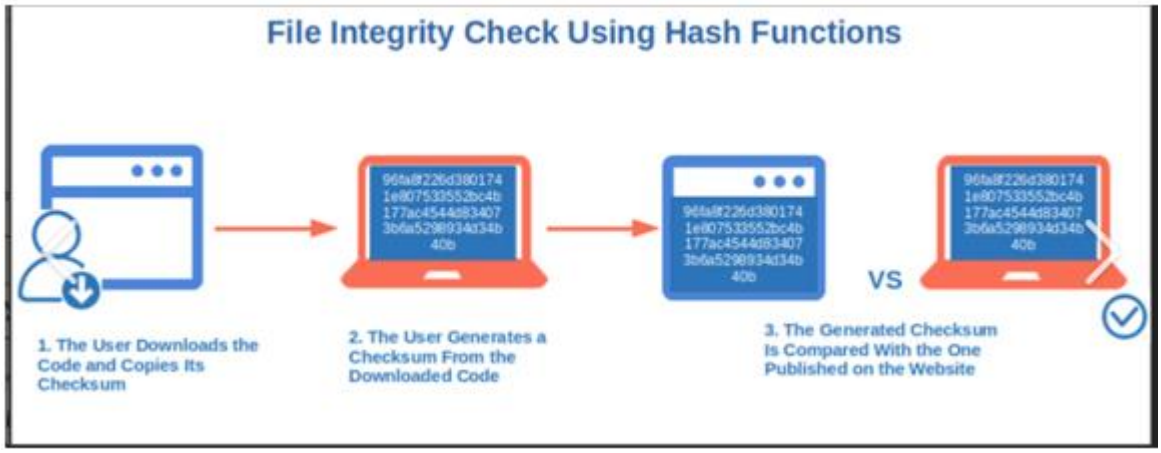
- Does not prevent attacks; it only detects changes
- Performance may reduce when monitoring a large number of files.
- Requires proper initial setup and secure storage of hash values.

EXPECTED OUTCOMES :

- **The system will successfully generate unique hash values for files using cryptographic hashing techniques**
- **It will detect any unauthorized changes, modifications, or deletions in monitored files**
- **The system will compare current and stored hash values to ensure file integrity.**
- **It will provide alerts or notification when file tempering is detected.**

- **The Project will demonstrate improved data security and reliability.**

  It will offer a simple and efficient solution for file monitoring.



APPLICATION USE IN ENGINEERING

- **The File Integrity Monitoring System using cryptographic hashing demonstrates the practical application of engineering principles and technical knowledge to ensure data security. The system is developed using programming tools such as Python, which enables efficient file handling and automation. Cryptographic hash functions like SHA-256 and MD5 are used to generate**

unique hash values for each file, ensuring that even a small change in the file can be detected. Engineering design principles such as modularity are applied by dividing the system into components like file scanning, hash generation, storage, and comparison, making the system organized and scalable. The implementation of hashing algorithms allows accurate and fast verification of file integrity by comparing current and previously stored hash values. Additionally, optimization techniques are used to improve performance, allowing the system to handle multiple files efficiently while minimizing resource usage and providing quick detection of any unauthorized modifications.

FUTURE WORK :

- The project can be further enhanced with the following improvements:

  ◈ Real-time Monitoring: Implement continuous real-time tracking of files

  ◈ Cloud Integration: Monitor files stored on cloud platforms

- Advanced Security: Use stronger hashing algorithms and encryption techniques
- Alert System: Add email/SMS notifications for instant alerts
- User Interface: Develop a graphical user interface (GUI) for ease of use
- AI Integration: Use machine learning to detect suspicious patterns or attacks
- Multi-System Support: Extend the system to monitor multiple devices in **a network .**

CONCLUSION :

The File Integrity Monitoring System using cryptographic hashing provides an effective solution for ensuring data security and detecting unauthorized

changes in files. By generating unique hash values for each file and comparing them over time, the system can easily identify any modification, deletion, or tampering. The project demonstrates that cryptographic hashing is a reliable and efficient technique for maintaining file integrity. It helps in protecting sensitive data from malware attacks, insider threats, and accidental changes. Overall, the system is simple, cost-effective, and useful for both individuals and organizations.

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